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Editorial Office
Journal of Sound and Vibration

Dear Editors,

Please consider our manuscript, “*Multiplication-Free Stochastic Semi-Discretization for the Efficient Moment Stability Analysis of Vibration Systems with Delay*” by Dániel Bachrathy and Henrik T. Sykora, for publication in the *Journal of Sound and Vibration* as a regular research paper.

The moment stability analysis of stochastic delay differential equations — the natural model class for machining dynamics, delayed feedback control, and connected vehicles under noise — has long been limited by the $\mathcal{O}(p^4)$ cost of assembling and multiplying the second-moment step operators of the semi-discretization framework. The manuscript removes this bottleneck and, we believe, changes what is practically computable in this field:

1. A **multiplication-free reformulation** evaluates the identical discrete second-moment operator in strictly $\mathcal{O}(p^2)$ time. Measured on the same machine and the same operator, the advantage over the explicit period product reaches 1.3×10^3 in wall-clock time and 6.3×10^3 in memory allocation already at $p = 192$, beyond which the classical route is infeasible while the new one continues to $p = 4096$.
2. A **Kronecker-factored Itô operator** removes the $\mathcal{O}(d^4)$ state-dimension wall: second-moment computations for systems with one hundred degrees of freedom — covariance eigenproblems with millions of unknowns — complete within minutes on a workstation, with a GPU port included in the accompanying open-source package.
3. **High-order collocation covariance blocks** raise the measured second-moment convergence order to $2S$ (rates consistent with order eight at $S = 4$), where all previous schemes of this family were limited to first order.
4. The methodology is demonstrated on a **two-degree-of-freedom milling process with spindle-speed variation**, time-varying delay, and multiplicative cutting-force noise. The resulting process map separates, for the first time at this fidelity, the deterministic stability gain of spindle-speed variation from the part that stochasticity takes back, and shows that the practically usable window is set by a surface-quality (stationary-variance) boundary lying strictly below all stability boundaries. The stationary variance is validated against an independent Monte-Carlo simulation to within 1.4%.

All results are reproducible: the methods are released in an open-source Julia package, every figure has a generating script, and the numerical experiments specify hardware, solver settings, and per-figure computational cost.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission and declare no competing interests. Given the topic, the readership of the *Journal of Sound and Vibration* — at the intersection of non-linear vibrations, machining dynamics, and computational methods — is the natural audience

for this work.

Thank you for your consideration.

Yours sincerely,

Dániel Bachrathy

(on behalf of the authors)

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